

AN ORDER-INTERPOLATION INEQUALITY FOR BESSEL FUNCTIONS

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ABSTRACT. We show that $J_{\mu+\nu}(r)^2 < J_{\nu-1/2}(r)^2 + J_{\nu+1/2}(r)^2$ holds whenever $\mu \in (-1/2, 1/2)$, $\nu \in [0, \infty)$, and $r \in (0, \infty)$. In fact, we prove a stronger version for any fixed non-trivial linear combination of the Bessel functions of the first and second kinds. This inequality can be regarded as a kind of interpolation with respect to order. As an application, we establish a dimension-comparison result for optimal constants of smoothing estimates for the free Schrödinger equation. Briefly, the optimal constant on $\mathbb{R}^{d+1}$ is at most twice that on $\mathbb{R}^d$ for each $d \geq 2$.

1. INTRODUCTION

It is classical and well known that the Bessel function of the first kind J_ν satisfies

$$rJ_\nu(r)^2 \leq \frac{2}{\pi} \quad (1.1)$$

for every $\nu \in [-1/2, 1/2]$ and $r \in (0, \infty)$, which was proved by Szegő [10, Equation (18)] (see also Szegő [11, Theorem 7.31.2]). Comparing this with Hankel's asymptotic formula [2, 10.17.3]

$$rJ_\nu(r)^2 = \frac{1}{\pi}(1 + \sin(2r - \nu\pi)) + O(r^{-1}), \quad r \rightarrow \infty \quad (1.2)$$

for each fixed $\nu \in \mathbb{R}$, we see that the constant $2/\pi$ in (1.1) is sharp. On the other hand, Szegő [10] also showed that (1.1) fails for $\nu \in \mathbb{R} \setminus [-1/2, 1/2]$, even though the asymptotic formula (1.2) is valid for every $\nu \in \mathbb{R}$. Indeed, we have

$$\begin{aligned} \nu \in (1/2, \infty) \cup \mathbb{Z}_{\leq -1} &\implies \frac{2}{\pi} < \sup_{r \in (0, \infty)} rJ_\nu(r)^2 < \infty, \\ \nu \in (-\infty, -1/2) \setminus \mathbb{Z}_{\leq -1} &\implies \sup_{r \in (0, \infty)} rJ_\nu(r)^2 = \infty. \end{aligned}$$

Moreover, Landau [6, Section 3.2] proved that the function

$$[1/2, \infty) \ni \nu \longmapsto \sup_{r \in (0, \infty)} rJ_\nu(r)^2$$

is strictly increasing and diverges to infinity as $\nu \rightarrow \infty$. Krasikov [5, Theorem 3] showed that

$$|r^2 - \nu^2 + 1/4|^{1/2} J_\nu(r)^2 < \frac{2}{\pi} \quad (1.3)$$

holds for every $\nu \in (1/2, \infty)$ and $r \in (0, \infty)$, which is a natural replacement of (1.1). The purpose of this paper is to give another generalization in terms of an interpolation inequality with respect to order. Notice that the inequality (1.1) can be rewritten as

$$J_\nu(r)^2 \leq J_{-1/2}(r)^2 + J_{1/2}(r)^2$$

by using

$$J_{-1/2}(r) = \left(\frac{2}{\pi r}\right)^{1/2} \cos r, \quad J_{1/2}(r) = \left(\frac{2}{\pi r}\right)^{1/2} \sin r.$$

This observation leads us to the following.

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Theorem 1.1. We define a family of functions $\{\mathbf{J}_\nu\}_{\nu \in \mathbb{R}}$ by

$$\mathbf{J}_\nu := aJ_\nu + bY_\nu$$

for fixed $(a, b) \in \mathbb{C}^2 \setminus \{(0, 0)\}$, where J_ν and Y_ν are the Bessel functions of the first and second kinds of order ν , respectively. Now let $\mu \in (-1/2, 1/2)$, $\nu \in [0, \infty)$, and $r \in (0, \infty)$. Then we have

$$|\mathbf{J}_{\mu+\nu}(r)|^2 < |\mathbf{J}_{\nu-1/2}(r)|^2 + |\mathbf{J}_{\nu+1/2}(r)|^2. \quad (1.4)$$

To the best of our knowledge, this simple-looking inequality does not seem to have appeared in the literature. See Section 2 for the proof. We remark that neither (1.3) nor (1.4) is stronger than the other. For example, (1.3) and (1.4) imply that

$$\frac{1}{2}\pi r J_1(r)^2 < \begin{cases} \frac{r}{|r^2 - 3/4|^{1/2}}, \\ \frac{1}{2}\pi r (J_{1/2}(r)^2 + J_{3/2}(r)^2) = 1 - \frac{\sin 2r}{r} + \frac{1 - \cos 2r}{2r^2}, \end{cases} \quad (1.5)$$

respectively. Then it is easy to see that

$$1 - \frac{\sin 2r}{r} + \frac{1 - \cos 2r}{2r^2} < \frac{r}{|r^2 - 3/4|^{1/2}}$$

holds when $r = \pi/4 + n\pi$, while the reverse inequality holds when $r = 3\pi/4 + n\pi$, where $n \in \mathbb{Z}_{\geq 0}$. Figure 1 shows the graphs of the functions appearing in (1.5).

In addition, we establish a dimension-comparison result for the free Schrödinger equation on $\mathbb{R}^d$ and $\mathbb{R}^{d+1}$ by using Theorem 1.1. Briefly, we have

$$\mathbf{C}^{(d+1)} \leq 2\mathbf{C}^{(d)}$$

for every $d \geq 2$, where $\mathbf{C}^{(d)}$ and $\mathbf{C}^{(d+1)}$ denote the optimal constants of a certain inequality on $\mathbb{R}^d$ and $\mathbb{R}^{d+1}$, respectively. See Section 3 for details.

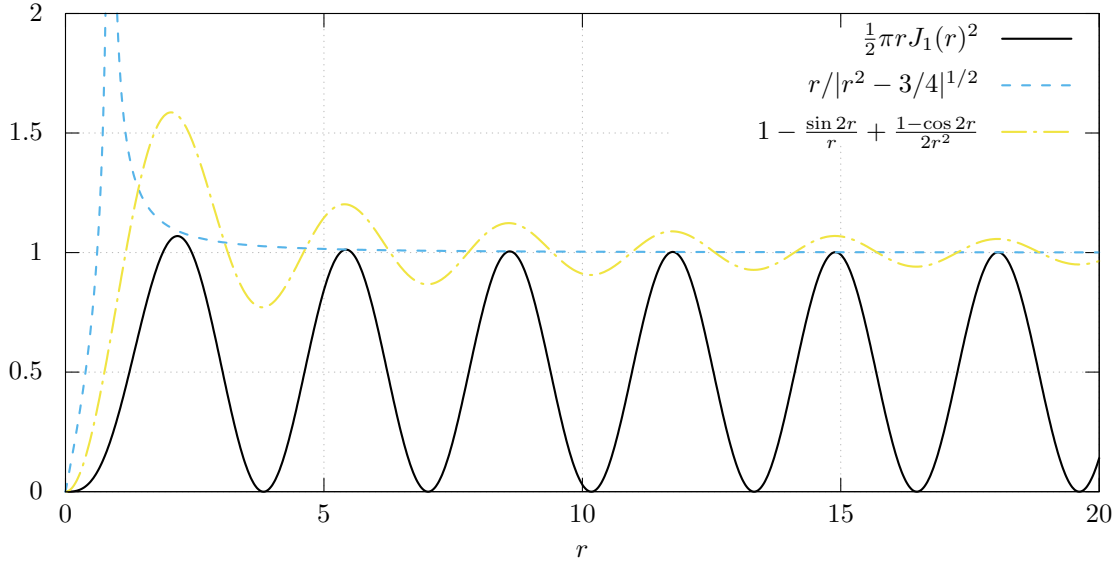


FIGURE 1. The graphs of the functions appearing in (1.5).

2. PROOF OF THEOREM 1.1

Our proof is based on three classical identities for the Bessel functions: the Wronskian identity, [Watson's](#) formula [13, Eq. (5) in Section 13.73], and [Nicholson's](#) formula [7, Eqs. (33), (34)]. For each $(\mu, \nu) \in \mathbb{R}^2$, we define $X_{\mu, \nu}: (0, \infty) \rightarrow \mathbb{R}$ by

$$X_{\mu, \nu}(r) := \det \begin{pmatrix} J_\mu & J_\nu \\ Y_\mu & Y_\nu \end{pmatrix} = J_\mu(r)Y_\nu(r) - J_\nu(r)Y_\mu(r),$$

that is, a cross product of J and Y . Then we have the following.

Proposition 2.1 (Wronskian identity). *Let $\nu \in \mathbb{R}$ and $r \in (0, \infty)$. Then we have*

$$X_{\nu+1/2, \nu-1/2}(r) = \frac{2}{\pi r}. \quad (2.1)$$

Proposition 2.2 ([Watson's](#) formula¹). *Let $\mu \in (-1, 1)$, $\nu \in \mathbb{R}$, and $r \in (0, \infty)$. Then we have*

$$X_{\mu+\nu, \nu}(r) = \frac{4 \sin(\mu\pi)}{\pi^2} \int_{s \in (0, \infty)} K_\mu(2r \sinh s) \exp(-(\mu + 2\nu)s) ds, \quad (2.2)$$

where K_μ is the modified Bessel function of the second kind of order μ .

Proposition 2.3 ([Nicholson's](#) formula). *Let $\nu \in \mathbb{R}$ and $r \in (0, \infty)$. Then we have*

$$J_\nu(r)^2 + Y_\nu(r)^2 = \frac{8}{\pi^2} \int_{s \in (0, \infty)} K_0(2r \sinh s) \cosh(2\nu s) ds. \quad (2.3)$$

Since K_ν is positive on $(0, \infty)$ for every $\nu \in \mathbb{R}$, we see that the identities (2.2) and (2.3) imply the following, respectively.

Corollary 2.4. *Let $\mu \in (0, 1)$ and $r \in (0, \infty)$. Then the function*

$$\nu \mapsto X_{\mu+\nu, \nu}(r)$$

is positive and strictly decreasing on $\mathbb{R}$.

Corollary 2.5. *Let $r \in (0, \infty)$. Then the function*

$$\nu \mapsto J_\nu(r)^2 + Y_\nu(r)^2$$

is even on $\mathbb{R}$ and strictly increasing on $[0, \infty)$.

Now we prove Theorem 1.1 by using Proposition 2.1 and Corollaries 2.4, 2.5.

Proof of Theorem 1.1. Using the Wronskian identity (2.1) and Cramer's rule, we obtain

$$\frac{2}{\pi r} \begin{pmatrix} J_{\mu+\nu}(r) \\ Y_{\mu+\nu}(r) \end{pmatrix} = X_{\nu+1/2, \mu+\nu}(r) \begin{pmatrix} J_{\nu-1/2}(r) \\ Y_{\nu-1/2}(r) \end{pmatrix} + X_{\mu+\nu, \nu-1/2}(r) \begin{pmatrix} J_{\nu+1/2}(r) \\ Y_{\nu+1/2}(r) \end{pmatrix},$$

so that

$$\frac{2}{\pi r} \mathbf{J}_{\mu+\nu}(r) = X_{\nu+1/2, \mu+\nu}(r) \mathbf{J}_{\nu-1/2}(r) + X_{\mu+\nu, \nu-1/2}(r) \mathbf{J}_{\nu+1/2}(r).$$

Thus, applying the Cauchy-Schwarz inequality, we get

$$\left(\frac{2}{\pi r} \right)^2 |\mathbf{J}_{\mu+\nu}(r)|^2 \leq (X_{\nu+1/2, \mu+\nu}(r)^2 + X_{\mu+\nu, \nu-1/2}(r)^2) (|\mathbf{J}_{\nu-1/2}(r)|^2 + |\mathbf{J}_{\nu+1/2}(r)|^2).$$

We also note that $X_{\nu+1/2, \nu-1/2}(r) \neq 0$ (which follows from the Wronskian identity (2.1)) and the assumption $(a, b) \neq (0, 0)$ imply that

$$|\mathbf{J}_{\nu-1/2}(r)|^2 + |\mathbf{J}_{\nu+1/2}(r)|^2 > 0.$$

¹Some references (e.g. Gradshteyn and Ryzhik [3, 6.617.1]) present [Watson's](#) formula (2.2) with the wrong sign in the exponential.

Therefore, it suffices to show that

$$X_{\nu+1/2,\mu+\nu}(r)^2 + X_{\mu+\nu,\nu-1/2}(r)^2 < \left(\frac{2}{\pi r}\right)^2$$

holds. Moreover, Corollary 2.4 gives us

$$\begin{aligned} 0 < X_{(1/2-\mu)+\mu+\nu,\mu+\nu}(r) &\leq X_{(1/2-\mu)+\mu,\mu}(r), \\ 0 < X_{\mu+\nu,\nu-1/2}(r) &\leq X_{\mu,-1/2}(r), \end{aligned}$$

so that it is reduced to

$$X_{1/2,\mu}(r)^2 + X_{\mu,-1/2}(r)^2 < \left(\frac{2}{\pi r}\right)^2.$$

Now observe that the left-hand side can be simplified as

$$X_{1/2,\mu}(r)^2 + X_{\mu,-1/2}(r)^2 = \frac{2}{\pi r} (J_\mu(r)^2 + Y_\mu(r)^2)$$

by substituting

$$J_{-1/2}(r) = -Y_{1/2}(r) = \left(\frac{2}{\pi r}\right)^{1/2} \cos r, \quad J_{1/2}(r) = Y_{-1/2}(r) = \left(\frac{2}{\pi r}\right)^{1/2} \sin r.$$

Then we get

$$J_\mu(r)^2 + Y_\mu(r)^2 < J_{1/2}(r)^2 + Y_{1/2}(r)^2 = \frac{2}{\pi r}$$

by Corollary 2.5, since $\mu \in (-1/2, 1/2)$. This completes the proof. $\square$

3. APPLICATION TO OPTIMAL CONSTANTS OF SMOOTHING ESTIMATES

Let $H_0 := -\Delta$ be the Schrödinger operator of a free particle on $\mathbb{R}^d$. Then it is well known that the so-called smoothing estimate

$$\int_{(x,t) \in \mathbb{R}^d \times \mathbb{R}} w(|x|) |\psi(H_0) e^{-itH_0} u_0(x)|^2 dx dt \leq C \|u_0\|_{L^2(\mathbb{R}^d)}^2 \quad (3.1)$$

holds for a suitable pair of functions $w, \psi: (0, \infty) \rightarrow [0, \infty)$, for example,

$$d \geq 3, \quad (w(r), \psi(s)) = ((1+r^2)^{-1}, (1+s)^{1/4}),$$

which is due to Kato and Yajima [4, Theorem 2]. The inequality (3.1) is usually referred to as a smoothing estimate. Hereinafter, we always assume that w and ψ are non-negative Borel measurable functions defined on $(0, \infty)$. Now let $C^{(d)}(w, \psi) \in [0, \infty]$ be the optimal constant of the inequality (3.1), that is,

$$C^{(d)}(w, \psi) := \sup_{u_0 \in L^2(\mathbb{R}^d) \setminus \{0\}} \frac{1}{\|u_0\|_{L^2(\mathbb{R}^d)}^2} \int_{(x,t) \in \mathbb{R}^d \times \mathbb{R}} w(|x|) |\psi(H_0) e^{-itH_0} u_0(x)|^2 dx dt.$$

Walther [12, Theorem 4.1] established the following formula for the optimal constant.

Theorem 3.1 ([12, Theorem 4.1]). *Let $w: (0, \infty) \rightarrow [0, \infty)$. For each $\nu \in \mathbb{R}$, we define $T_\nu w: (0, \infty) \rightarrow [0, \infty]$ by*

$$T_\nu w(s) := \pi \int_{r \in (0, \infty)} rs J_\nu(rs)^2 w(r) dr. \quad (3.2)$$

Then, for every $d \geq 2$ and $\psi: (0, \infty) \rightarrow [0, \infty)$, we have

$$C^{(d)}(w, \psi) = \sup_{k \in \mathbb{Z}_{\geq 0}} \operatorname{ess\,sup}_{s \in (0, \infty)} s^{-1} \psi(s^2)^2 T_{k+d/2-1} w(s). \quad (3.3)$$

Notice that [Walther's](#) formula immediately implies

$$\mathbf{C}^{(d+2)}(w, \psi) \leq \mathbf{C}^{(d)}(w, \psi)$$

for every $d \geq 2$ and $w, \psi: (0, \infty) \rightarrow [0, \infty)$, since

$$\mathbf{C}^{(d+2)}(w, \psi) = \sup_{k \in \mathbb{Z}_{\geq 0}} \operatorname{ess\,sup}_{s \in (0, \infty)} s^{-1} \psi(s^2)^2 T_{k+(d+2)/2-1} w(s) = \sup_{k \in \mathbb{Z}_{\geq 1}} \operatorname{ess\,sup}_{s \in (0, \infty)} s^{-1} \psi(s^2)^2 T_{k+d/2-1} w(s).$$

Therefore, it is natural to ask if we have

$$\mathbf{C}^{(d+1)}(w, \psi) \lesssim \mathbf{C}^{(d)}(w, \psi).$$

Our Theorem 1.1 gives an affirmative answer with an explicit constant. In fact, Theorem 1.1 implies

$$T_\nu w(s) \leq T_{\nu-1/2} w(s) + T_{\nu+1/2} w(s) \leq 2 \max\{T_{\nu-1/2} w(s), T_{\nu+1/2} w(s)\}$$

for every $w: (0, \infty) \rightarrow [0, \infty)$, $\nu \in [0, \infty)$, and $s \in (0, \infty)$. Combining this inequality with [Walther's](#) formula (3.3), we obtain

$$\mathbf{C}^{(d+1)}(w, \psi) \leq 2\mathbf{C}^{(d)}(w, \psi). \quad (3.4)$$

A related refinement is obtained by comparing these constants with the optimal constant for the Dirac smoothing estimate. Let $H_{D,m}$ be the Dirac operator of a free particle with mass $m \in [0, \infty)$ on $\mathbb{R}^d$, and let $\mathbf{C}_{D,m}^{(d)}(w, \psi) \in [0, \infty]$ be the optimal constant of the inequality

$$\int_{(x,t) \in \mathbb{R}^d \times \mathbb{R}} w(|x|) |H_{D,m}|^{-1/2} \psi(H_0) e^{-itH_{D,m}} u_0(x) |^2 dx dt \leq C \|u_0\|_{L^2(\mathbb{R}^d, \mathbb{C}^N)}^2,$$

where $N := 2^{\lfloor (d+1)/2 \rfloor}$. Suzuki [8] showed the following analogue of Theorem 3.1.

Theorem 3.2 ([8, Theorem 1.5]). *Let $m \in [0, \infty)$ and $w: (0, \infty) \rightarrow [0, \infty)$. For each $\nu \in \mathbb{R}$, we define $\tilde{T}_{\nu,m} w: (0, \infty) \rightarrow [0, \infty]$ by*

$$\tilde{T}_{\nu,m} w(s) := T_\nu w(s) + T_{\nu+1} w(s) + \frac{m}{(m^2 + s^2)^{1/2}} |T_\nu w(s) - T_{\nu+1} w(s)|, \quad (3.5)$$

where $T_\nu w$ is as in (3.2). Then, for every $d \geq 2$ and $\psi: (0, \infty) \rightarrow [0, \infty)$, we have

$$\mathbf{C}_{D,m}^{(d)}(w, \psi) = \sup_{k \in \mathbb{Z}_{\geq 0}} \operatorname{ess\,sup}_{s \in (0, \infty)} s^{-1} \psi(s^2)^2 \tilde{T}_{k+d/2-1,m} w(s).$$

Combining (1.4) and (3.5), we see that

$$\max\{T_{\nu-1/2} w(s), T_\nu w(s)\} \leq \tilde{T}_{\nu-1/2,m} w(s) \leq 2 \max\{T_{\nu-1/2} w(s), T_{\nu+1/2} w(s)\}$$

for every $w: (0, \infty) \rightarrow [0, \infty)$, $m \in [0, \infty)$, $\nu \in [0, \infty)$, and $s \in (0, \infty)$. Hence, we obtain the following:

Theorem 3.3. *We have*

$$\max\{\mathbf{C}^{(d)}(w, \psi), \mathbf{C}^{(d+1)}(w, \psi)\} \leq \mathbf{C}_{D,m}^{(d)}(w, \psi) \leq 2\mathbf{C}^{(d)}(w, \psi) \quad (3.6)$$

for every $d \geq 2$, $m \in [0, \infty)$, and $w, \psi: (0, \infty) \rightarrow [0, \infty)$.

We note that the inequality (3.6) is sharp for every $d \geq 2$ in the following sense: we have

$$\lim_{p \uparrow d} \frac{\mathbf{C}_{D,0}^{(d)}(w_p, \psi_p)}{\max\{\mathbf{C}^{(d)}(w_p, \psi_p), \mathbf{C}^{(d+1)}(w_p, \psi_p)\}} = 1,$$

$$\lim_{p \downarrow 1} \frac{\mathbf{C}_{D,0}^{(d)}(w_p, \psi_p)}{\mathbf{C}^{(d)}(w_p, \psi_p)} = 2$$

for $(w_p(r), \psi_p(s)) = (r^{-p}, s^{(2-p)/4})$; see [1, Theorem 1.6] and Suzuki [8, Theorem 1.8] for details. Meanwhile, as of this writing, the sharpness of the inequality (3.4) remains open. Indeed, we do not know if there exists (w, ψ) such that

$$\mathbf{C}^{(d)}(w, \psi) < \mathbf{C}^{(d+1)}(w, \psi) \leq 2\mathbf{C}^{(d)}(w, \psi).$$

We remark that the following sufficient condition for

$$\mathbf{C}^{(d+1)}(w, \psi) \leq \mathbf{C}^{(d)}(w, \psi) \quad (3.7)$$

was recently obtained by Suzuki [9].

Theorem 3.4 ([9, Theorem 1.8]). *Let $w: (0, \infty) \rightarrow [0, \infty)$ be such that*

$$w(r) = \int_{a \in (0, \infty)} \exp(-ar^2) d\lambda(a)$$

for some non-negative Borel measure λ on $(0, \infty)$. Then the function

$$[0, \infty) \ni \nu \mapsto T_\nu w(s) \in [0, \infty]$$

is non-increasing for each $s \in (0, \infty)$. Consequently, the inequality (3.7) holds for every $d \geq 2$ and $\psi: (0, \infty) \rightarrow [0, \infty)$.

For example, $w(r) = (1 + r^2)^{-p/2}$ satisfies the assumption of Theorem 3.4 for every $p \in (0, \infty)$, since

$$(1 + r^2)^{-p/2} = \frac{1}{\Gamma(p/2)} \int_{a \in (0, \infty)} \exp(-ar^2) a^{p/2-1} \exp(-a) da.$$

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$$J_\nu(r)^2 \leq J_{\nu-1/2}(r)^2 + J_{\nu+1/2}(r)^2$$

in order to establish Theorem 3.3. The author would like to thank Timothy Budd for suggesting the more general inequality

$$J_{\mu+\nu}(r)^2 \leq J_{\nu-1/2}(r)^2 + J_{\nu+1/2}(r)^2$$

on MathOverflow².

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